Midterm – MATH 518 – Fall 2025

DATE: Monday, October 20 TIME: 4:05 – 5:25

EXAMINER: Romain Branchereau TOTAL: 40 points

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

- There are four problems, each worth 10 points, for a total of 40 points.
- Notes, calculators and other electronic devices are not permitted.
- Write your solutions directly on the booklet, under the question (you can use the back of the page).
- You can use any result stated in class or that appeared on the exercise sheets.
- The last two pages can be used as scrap paper and will not be graded.
- Throughout the exam, k is a field of characteristic 0.

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1. In the following two cases, describe the algebraic set V, determine the ideal $I(V)$ and if V is irreducible .
   (a) (5 points) The affine algebraic set $V = V(x^2y^3) \subset k^2$.
   (b) (5 points) The projective algebraic set $V = V(f, g) \subset \mathbb{P}^2(k)$, where $f = x^2+y^2-z^2$ and $g = x - z$ in $k[x, y, z]$.

2. (10 points) Find the decomposition of $V = V(y^2 - xz, z^2 - y^3) \subset k^3$ into irreducibles.

3. Let $\varphi\colon V \longrightarrow W$ be a morphism between two affine varieties $V \subset k^n$ and $W \subset k^m$, and $\varphi^*\colon k[W] \longrightarrow k[V]$ the induced k-algebra morphism.
   (a) (5 points) Show that $I(\varphi(V)) = \ker(\varphi^*)$.
   (b) (5 points) Deduce that φ^* is injective, if and only if $\varphi(V)$ is dense in W. *Such a map is called a dominant map.*

4. (10 points) Show that the affine line k and the hyperbola $V = V(xy - 1) \subset k^2$ are not ismorphic as affine varieties.

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